

KKT Second-Order Conditions

The stationarity equation in the KKT first-order conditions is a balance of forces. A balanced particle may sit at the bottom of a bowl, at the top of a hill, or on a saddle. Curvature distinguishes these possibilities, but a constrained problem does not expose every ambient direction to the particle. Curvature must be examined only along feasible motions that have not already been classified by first-order information.

Learning objectives

After this chapter, you should be able to:

- distinguish the KKT second-order necessary condition as a filter from the KKT second-order sufficient condition as a certificate;
- define the critical cone first from $\nabla f(\mathbf{x}^*)$ and then derive its two equivalent representations from the KKT first-order conditions;
- explain mechanically why loaded active inequalities become equalities in the critical cone while unloaded active inequalities do not;
- derive the Lagrangian Hessian by eliminating the acceleration of a feasible path;
- state, prove, and apply the KKT second-order necessary and sufficient conditions (SONC and SOSC) with their correct assumptions;
- recognize when the critical cone is a subspace and use a reduced Hessian;
- interpret zero critical curvature as a genuinely inconclusive higher-order case.

13.1 Why constrained curvature is different

Throughout this chapter, we use the smooth nonlinear program and the convention $h_j \leq 0$ from Equation (12.2). The problem data are assumed to be twice continuously differentiable near the point under discussion. The active set, Lagrangian, KKT point, and KKT triple are those defined in Definitions 12.2 to 12.4. In particular, the KKT FONC in Theorem 12.10 supplies the first-order candidates examined here; its four components are Equations (12.7) to (12.10).

For an unconstrained problem, every sufficiently small displacement is feasible, so a second-order test examines the objective Hessian in every direction. Under constraints, local optimality compares a candidate only with nearby feasible points. Negative ambient curvature in a direction that immediately leaves the feasible set therefore does not rule out constrained local minimality, as illustrated in Figure 13.1.

In the figure, the origin is the unique global minimizer of the restriction $f(0, x_2) = x_2^2$, although $\mathbf{H}_f(\mathbf{0}) = \text{diag}(-2, 2)$ is indefinite: its negative-curvature directions are infeasible. Among feasible directions, only those with zero first-order objective change still need a second-

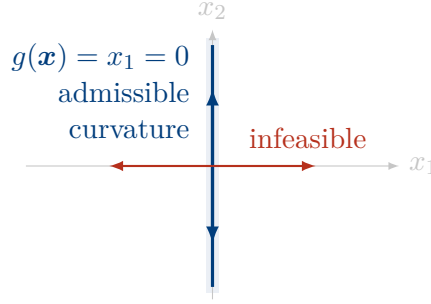


Figure 13.1: Under constraints, curvature is tested only along the relevant feasible directions. For $f(x_1, x_2) = -x_1^2 + x_2^2$ with $x_1 = 0$, the negative ambient curvature occurs only in infeasible directions.

order test, and a curved constraint can itself contribute curvature. The critical cone isolates the former directions, while the Lagrangian Hessian records both objective and constraint curvature.

13.2 The critical cone

Recall from Chapter 12 that the linearized feasible cone is

$$T_{\text{lin}}(\mathbf{x}^*) := \left\{ \mathbf{d} : \begin{array}{ll} \nabla g_i(\mathbf{x}^*)^\top \mathbf{d} = 0, & i = 1, \dots, m, \\ \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} \leq 0, & j \in J(\mathbf{x}^*) \end{array} \right\}. \quad (13.1)$$

Under LICQ, $T_{\text{lin}}(\mathbf{x}^*) = T_{\mathcal{S}}(\mathbf{x}^*)$ by the cone identity established in Chapter 12. The definition below still makes sense without LICQ, but then a linearized direction need not be an actual tangent direction.

Definition 13.1 (Critical cone). The critical cone at a feasible point $\mathbf{x}^*$ is

$$C_{\text{crit}}(\mathbf{x}^*) := \{\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*) : \nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0\}. \quad (13.2)$$

This definition uses only the problem data and the point. It contains the linearized feasible directions along which first-order objective information is silent. At a point satisfying the KKT first-order conditions, directions in $T_{\text{lin}}(\mathbf{x}^*)$ cannot have negative first-order change, and the critical cone is the zero-change part.

Proposition 13.1 (KKT representations of the critical cone). *Let $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ be a KKT triple. Define*

$$J_+ := \{j \in J(\mathbf{x}^*) : \nu_j^* > 0\}, \quad J_0 := \{j \in J(\mathbf{x}^*) : \nu_j^* = 0\}.$$

Then

$$C_{\text{crit}}(\mathbf{x}^*) = \left\{ \mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*) : \nu_j^* \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0 \text{ for every } j \in J(\mathbf{x}^*) \right\} \quad (13.3)$$

$$= \left\{ \mathbf{d} : \begin{array}{ll} \nabla g_i(\mathbf{x}^*)^\top \mathbf{d} = 0, & i = 1, \dots, m, \\ \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0, & j \in J_+, \\ \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} \leq 0, & j \in J_0 \end{array} \right\}. \quad (13.4)$$

Proof. For $\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*)$, KKT first-order stationarity Equation (12.7) and equality tangency give

$$\nabla f(\mathbf{x}^*)^\top \mathbf{d} = - \sum_{j \in J(\mathbf{x}^*)} \nu_j^* \nabla h_j(\mathbf{x}^*)^\top \mathbf{d}. \quad (13.5)$$

Each term $-\nu_j^* \nabla h_j(\mathbf{x}^*)^\top \mathbf{d}$ on the right is nonnegative because $\nu_j^* \geq 0$ and $\nabla h_j(\mathbf{x}^*)^\top \mathbf{d} \leq 0$. Thus $\nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0$ exactly when their sum is zero, which holds exactly when every term is zero. This proves Equation (13.3). For $j \in J_+$, the multiplier is positive, so the product condition reduces to $\nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0$ for $j \in J_+$, while the indices in J_0 impose no condition beyond their original linearized inequalities. This is Equation (13.4). $\square$

Important note: Loaded and unloaded walls

A direction in the critical cone does zero first-order objective work. A wall with $\nu_j^* > 0$ carries load, so zero work forces the motion to be tangent to that wall: $\nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0$. An active wall with $\nu_j^* = 0$ carries no load and may be left by moving into its feasible side: $\nabla h_j(\mathbf{x}^*)^\top \mathbf{d} \leq 0$.

Warning

The conceptual definition Equation (13.2) is independent of a choice of KKT multipliers. When LICQ fails, the multiplier representation may look different for different multiplier pairs but must describe the same set. By contrast, the Lagrangian Hessian introduced later can genuinely depend on the selected multiplier pair.

Example 13.1 (A loaded wall, an unloaded wall, and a genuine cone). Consider

$$\min_{x_1, x_2} f(\mathbf{x}) = x_1 + x_2^2 \quad \text{subject to} \quad h_1(\mathbf{x}) = -x_1 \leq 0, \quad h_2(\mathbf{x}) = -x_2 \leq 0. \quad (13.6)$$

The feasible set is the first quadrant, and $f(\mathbf{x}) \geq 0$ there with equality only at $\mathbf{x}^* = \mathbf{0}$. Thus the origin is the unique global minimizer. Stationarity is

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \nu_1^* \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \nu_2^* \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \mathbf{0},$$

so $(\nu_1^*, \nu_2^*) = (1, 0)$, $J_+ = \{1\}$, and $J_0 = \{2\}$. Hence

$$T_{\text{lin}}(\mathbf{0}) = \{\mathbf{d} : d_1 \geq 0, d_2 \geq 0\}, \quad C_{\text{crit}}(\mathbf{0}) = \{(0, d_2)^\top : d_2 \geq 0\}.$$

The critical cone is the nonnegative vertical ray, not a linear subspace. See Figure 13.2 for an illustration.

13.3 The Lagrangian Hessian as the relevant curvature

13.3.1 Realizing critical directions by controlled feasible paths

An arbitrary tangent sequence need not possess a controlled second derivative. Under LICQ, the inverse function theorem supplies the smooth path required for the second-order calculation.

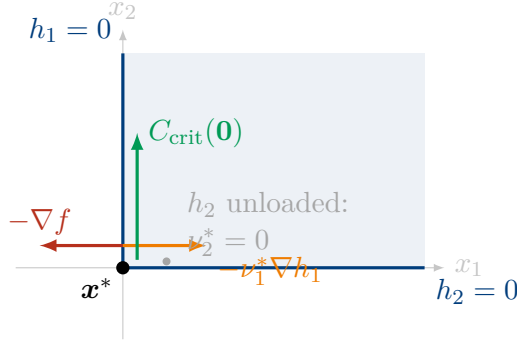


Figure 13.2: Critical cone with a loaded wall and an unloaded wall.

Lemma 13.2 (Linearized-direction realization under LICQ). *Assume all constraints are twice continuously differentiable near $\mathbf{x}^*$ and LICQ holds there. For $\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*)$, define*

$$J(\mathbf{d}) := \{j \in J(\mathbf{x}^*) : \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0\}.$$

Then there exist $\varepsilon > 0$ and a twice continuously differentiable curve $\mathbf{x} : [0, \varepsilon) \rightarrow \mathcal{S}$ such that

$$\mathbf{x}(0) = \mathbf{x}^*, \quad \mathbf{x}'(0) = \mathbf{d},$$

and, for all sufficiently small $t \geq 0$,

$$g_i(\mathbf{x}(t)) = 0 \quad (i = 1, \dots, m), \quad h_j(\mathbf{x}(t)) = 0 \quad (j \in J(\mathbf{d})).$$

Proof. Write $J(\mathbf{d}) = \{j_1, \dots, j_s\}$ and define

$$M := [\nabla g_1(\mathbf{x}^*) \ \cdots \ \nabla g_m(\mathbf{x}^*) \ \nabla h_{j_1}(\mathbf{x}^*) \ \cdots \ \nabla h_{j_s}(\mathbf{x}^*)].$$

LICQ for the full active family implies that M has full column rank. Let the columns of $\mathbf{Z}_d$ form a basis of $\text{Null}(M^\top)$. Then the square matrix whose row blocks are $M^\top$ and $\mathbf{Z}_d^\top$ is nonsingular. Define

$$\Phi(\mathbf{z}) := (g_1(\mathbf{z}), \dots, g_m(\mathbf{z}), h_{j_1}(\mathbf{z}), \dots, h_{j_s}(\mathbf{z}), \mathbf{Z}_d^\top(\mathbf{z} - \mathbf{x}^*)).$$

The derivative of Φ at $\mathbf{x}^*$ is nonsingular, so the inverse function theorem gives a C^2 local inverse. Define $\mathbf{x}(t)$ near zero by

$$\Phi(\mathbf{x}(t)) = (\mathbf{0}, \mathbf{0}, t\mathbf{Z}_d^\top \mathbf{d}).$$

The first two blocks keep the selected constraints at equality. Differentiating at $t = 0$ gives

$$\begin{pmatrix} M^\top \\ \mathbf{Z}_d^\top \end{pmatrix} \mathbf{x}'(0) = \begin{pmatrix} \mathbf{0} \\ \mathbf{Z}_d^\top \mathbf{d} \end{pmatrix} = \begin{pmatrix} M^\top \\ \mathbf{Z}_d^\top \end{pmatrix} \mathbf{d},$$

because $M^\top \mathbf{d} = \mathbf{0}$ by the definition of $J(\mathbf{d})$ and linearized equality feasibility. Nonsingularity therefore yields $\mathbf{x}'(0) = \mathbf{d}$.

If $j \in J(\mathbf{x}^*) \setminus J(\mathbf{d})$, then $\nabla h_j(\mathbf{x}^*)^\top \mathbf{d} < 0$, so

$$h_j(\mathbf{x}(t)) = t\nabla h_j(\mathbf{x}^*)^\top \mathbf{d} + o(t) < 0$$

for all sufficiently small $t > 0$. Inactive inequalities remain strict by continuity. Reducing ε if necessary makes the entire curve feasible. $\square$

Warning

A given differentiable feasible curve yields a direction in $T_{\text{lin}}(\mathbf{x}^*)$ without LICQ. LICQ is used in the converse direction: it realizes *every* desired linearized direction by a controlled C^2 feasible path. In the SONC proof, the path must keep all constraints in $J(\mathbf{d})$ —and therefore all loaded constraints J_+ —active. An arbitrary feasible curve does not provide the equality needed below.

13.3.2 Eliminating the path acceleration

For the Lagrangian defined in [Definition 12.3](#), define its Hessian at a KKT triple by

$$\mathbf{H}_L := \nabla_{xx}^2 L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{H}_f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \mathbf{H}_{g_i}(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \mathbf{H}_{h_j}(\mathbf{x}^*). \quad (13.7)$$

Lemma 13.3 (Second-order feasibility equations). *Suppose*

$$\mathbf{x}(t) = \mathbf{x}^* + t\mathbf{d} + \frac{1}{2}t^2\mathbf{w} + o(t^2).$$

If $g_i(\mathbf{x}(t)) = 0$ for all small t , then

$$\nabla g_i(\mathbf{x}^*)^\top \mathbf{w} + \mathbf{d}^\top \mathbf{H}_{g_i}(\mathbf{x}^*) \mathbf{d} = 0. \quad (13.8)$$

If $j \in J(\mathbf{x}^)$, $\nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0$, and $h_j(\mathbf{x}(t)) \leq 0$ for all sufficiently small $t \geq 0$, then*

$$\nabla h_j(\mathbf{x}^*)^\top \mathbf{w} + \mathbf{d}^\top \mathbf{H}_{h_j}(\mathbf{x}^*) \mathbf{d} \leq 0. \quad (13.9)$$

If in addition $h_j(\mathbf{x}(t)) = 0$ for all small t , equality holds in [Equation \(13.9\)](#).

Proof. Taylor expansion gives

$$\begin{aligned} g_i(\mathbf{x}(t)) &= g_i(\mathbf{x}^*) + t \nabla g_i(\mathbf{x}^*)^\top \mathbf{d} \\ &\quad + \frac{t^2}{2} \left(\nabla g_i(\mathbf{x}^*)^\top \mathbf{w} + \mathbf{d}^\top \mathbf{H}_{g_i}(\mathbf{x}^*) \mathbf{d} \right) + o(t^2). \end{aligned}$$

The constant and linear terms vanish. Divide by $t^2/2$ and let $t \downarrow 0$ to obtain [Equation \(13.8\)](#). The identical expansion for an active inequality with zero linear term, together with $h_j(\mathbf{x}(t)) \leq 0$, gives [Equation \(13.9\)](#); if the constraint remains an equality, the same argument applied to both signs gives equality. $\square$

Proposition 13.4 (Curvature along a controlled feasible path). *Let $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ be a KKT triple. Let $\mathbf{x}(t)$ be twice differentiable with $\mathbf{x}(0) = \mathbf{x}^*$, $\mathbf{x}'(0) = \mathbf{d}$, and $\mathbf{x}(t) \in \mathcal{S}$ for all sufficiently small $t \geq 0$. Suppose all equalities and all positive-multiplier active inequalities remain equalities along the path. Then*

$$\left. \frac{d^2}{dt^2} f(\mathbf{x}(t)) \right|_{t=0} = \mathbf{d}^\top \mathbf{H}_L \mathbf{d}. \quad (13.10)$$

Proof. Let $\mathbf{w} = \mathbf{x}''(0)$. Twice differentiating the objective gives

$$(f \circ \mathbf{x})''(0) = \mathbf{d}^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \nabla f(\mathbf{x}^*)^\top \mathbf{w}.$$

KKT first-order stationarity Equation (12.7) gives

$$\nabla f(\mathbf{x}^*)^\top \mathbf{w} = - \sum_i \lambda_i^* \nabla g_i(\mathbf{x}^*)^\top \mathbf{w} - \sum_{j \in J_+} \nu_j^* \nabla h_j(\mathbf{x}^*)^\top \mathbf{w},$$

because inactive and zero-multiplier inequalities contribute nothing. By Lemma 13.3, the constraints that remain equalities satisfy

$$-\nabla g_i(\mathbf{x}^*)^\top \mathbf{w} = \mathbf{d}^\top \mathbf{H}_{g_i}(\mathbf{x}^*) \mathbf{d}, \quad -\nabla h_j(\mathbf{x}^*)^\top \mathbf{w} = \mathbf{d}^\top \mathbf{H}_{h_j}(\mathbf{x}^*) \mathbf{d}$$

for $j \in J_+$. Substitution gives exactly Equation (13.7) inside the quadratic form, proving Equation (13.10). $\square$

Example 13.2 (A flat objective whose feasible restriction curves upward). Consider

$$\min f(\mathbf{x}) = x_2 \quad \text{subject to} \quad g(\mathbf{x}) = x_2 - x_1^2 = 0.$$

At $\mathbf{x}^* = \mathbf{0}$, stationarity gives $\lambda^* = -1$. Although $\mathbf{H}_f(\mathbf{0}) = 0$,

$$\mathbf{H}_L = -\mathbf{H}_g(\mathbf{0}) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

The tangent directions are $\mathbf{d} = (d_1, 0)^\top$, and the feasible curve $\mathbf{x}(t) = (t, t^2)^\top$ satisfies

$$f(\mathbf{x}(t)) = t^2, \quad (f \circ \mathbf{x})''(0) = 2 = \begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{H}_L \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Thus the constraint curvature, weighted by the multiplier, supplies all of the effective second-order growth. For an illustration, see Figure 13.3.

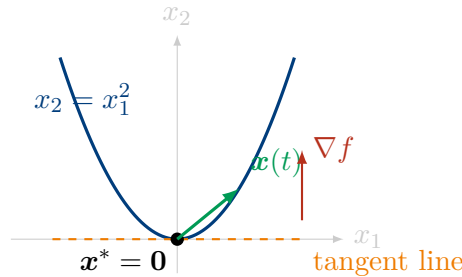


Figure 13.3: Although $f(\mathbf{x}) = x_2$ is flat to second order in the ambient space, the curved feasible path bends upward. The multiplier-weighted constraint curvature is exactly the missing contribution recorded by $\mathbf{H}_L$.

13.4 KKT second-order necessity

Theorem 13.5 (KKT second-order necessary condition (SONC)). *Suppose $\mathbf{x}^*$ is a local minimizer of Equation (12.2), the problem data are twice continuously differentiable, and LICQ holds at $\mathbf{x}^*$. Let $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ be the unique KKT multiplier pair. Then*

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} \geq 0 \quad \text{for every } \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*). \quad (13.11)$$

Proof. Fix $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$ and define $J(\mathbf{d})$ as in Lemma 13.2. By Proposition 13.1, $J_+ \subseteq J(\mathbf{d})$. The realization lemma supplies a twice differentiable feasible curve $\mathbf{x}(t)$ with $\mathbf{x}(0) = \mathbf{x}^*$, $\mathbf{x}'(0) = \mathbf{d}$, and every constraint in $J(\mathbf{d})$ kept active.

Set $\phi(t) = f(\mathbf{x}(t))$. Local minimality gives $\phi(t) \geq \phi(0)$ for all sufficiently small $t \geq 0$, while criticality gives

$$\phi'(0) = \nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0.$$

Taylor expansion therefore yields

$$0 \leq \lim_{t \downarrow 0} \frac{2(\phi(t) - \phi(0) - t\phi'(0))}{t^2} = \phi''(0).$$

Every positive-multiplier inequality remains active along the chosen path, so Proposition 13.4 gives $\phi''(0) = \mathbf{d}^\top \mathbf{H}_L \mathbf{d}$. This proves Equation (13.11). $\square$

Corollary 13.6 (Unconstrained and equality-constrained special cases). *Under the hypotheses of Theorem 13.5, the unconstrained SONC reduces to $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\mathbf{H}_f(\mathbf{x}^*) \succeq \mathbf{0}$. With equality constraints only and linearly independent equality gradients, SONC is*

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} \geq 0 \quad \text{whenever} \quad \nabla g_i(\mathbf{x}^*)^\top \mathbf{d} = 0 \quad (i = 1, \dots, m).$$

Proof. With no constraints, the KKT FONC first gives $\nabla f(\mathbf{x}^*) = \mathbf{0}$. Hence $T_{\text{lin}}(\mathbf{x}^*) = C_{\text{crit}}(\mathbf{x}^*) = \mathbb{R}^n$ and $\mathbf{H}_L = \mathbf{H}_f(\mathbf{x}^*)$, so the first statement is Theorem 13.5. With equality constraints only, KKT first-order stationarity makes $\nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0$ on the null space of all equality gradients. Hence that null space is the entire critical cone, and the second statement again follows from Theorem 13.5. $\square$

Important note: Workflow for the KKT second-order necessary condition

For a feasible candidate:

1. verify LICQ at the candidate, solve the KKT first-order equations, and identify J_+ and J_0 ;
2. construct $C_{\text{crit}}(\mathbf{x}^*)$ and form $\mathbf{H}_L$;
3. if $C_{\text{crit}}(\mathbf{x}^*)$ is a subspace, let the columns of $\mathbf{Z}$ form a basis so that $C_{\text{crit}}(\mathbf{x}^*) = \text{Range}(\mathbf{Z})$, and test the reduced Hessian $\mathbf{Z}^\top \mathbf{H}_L \mathbf{Z}$; otherwise test the quadratic form directly on the cone;
4. determine whether $\mathbf{d}^\top \mathbf{H}_L \mathbf{d} \geq 0$ on the entire critical cone and, in particular, search for a violating direction $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$ with $\mathbf{d}^\top \mathbf{H}_L \mathbf{d} < 0$.

Under the SONC hypotheses, one negative critical direction rejects the candidate. If nonnegativity is established on the entire cone, SONC is passed but local minimality is still not proved. Merely failing to locate a negative direction computationally establishes neither statement. The reduced-Hessian reduction in Step 3 is justified in the

next section.

13.4.1 Critical subspaces and reduced Hessians

Recall from [Definition 12.5](#) that strict complementarity holds at a KKT triple precisely when

$$\nu_j^* > 0 \quad \text{for every } j \in J(\mathbf{x}^*).$$

Proposition 13.7 (When the critical cone is a subspace). *Let $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ be a fixed KKT triple. The critical cone at $\mathbf{x}^*$ is a linear subspace in either of the following cases:*

1. *only equality constraints are present;*
2. *this KKT triple satisfies strict complementarity.*

In the second case,

$$C_{\text{crit}}(\mathbf{x}^*) = \{\mathbf{d} : \nabla g_i(\mathbf{x}^*)^\top \mathbf{d} = 0, \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0 \ (j \in J(\mathbf{x}^*))\}.$$

Proof. With equality constraints only, the linearized cone is the null space of the equality gradients, and KKT stationarity makes the objective gradient orthogonal to that null space. Hence the whole linearized cone is critical. Under strict complementarity, $J_+ = J(\mathbf{x}^*)$ and $J_0 = \emptyset$. Substitution into [Equation \(13.4\)](#) leaves only homogeneous linear equalities. Their solution set is a subspace. $\square$

Warning

Strict complementarity and LICQ are different assumptions. Strict complementarity concerns multiplier signs; LICQ concerns linear independence of constraint gradients. Neither definition contains the other. The subspace identity above is algebraic, whereas applying SONC as a necessary condition still requires the regularity assumption in [Theorem 13.5](#).

Let $\mathbf{A}$ collect the equality gradients and, under strict complementarity, all active inequality gradients. Let the columns of $\mathbf{Z}$ form a basis of $\text{Null}(\mathbf{A}^\top) = C_{\text{crit}}(\mathbf{x}^*)$.

Proposition 13.8 (Reduced-Hessian test). *If $C_{\text{crit}}(\mathbf{x}^*) = \text{Range}(\mathbf{Z})$, then*

$$\begin{aligned} \mathbf{d}^\top \mathbf{H}_L \mathbf{d} &\geq 0 \quad \forall \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*) \iff \mathbf{Z}^\top \mathbf{H}_L \mathbf{Z} \succeq 0, \\ \mathbf{d}^\top \mathbf{H}_L \mathbf{d} &> 0 \quad \forall \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*) \setminus \{\mathbf{0}\} \iff \mathbf{Z}^\top \mathbf{H}_L \mathbf{Z} \succ 0. \end{aligned}$$

Proof. Every critical direction has the unique form $\mathbf{d} = \mathbf{Z}\mathbf{y}$, because the columns of $\mathbf{Z}$ are a basis. Therefore

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} = \mathbf{y}^\top (\mathbf{Z}^\top \mathbf{H}_L \mathbf{Z}) \mathbf{y}.$$

Nonnegativity for every $\mathbf{d}$ is precisely positive semidefiniteness of the reduced matrix. Since $\mathbf{Z}$ has full column rank, $\mathbf{d} \neq \mathbf{0}$ exactly when $\mathbf{y} \neq \mathbf{0}$, proving the strict version. $\square$

Proposition 13.9 (Independence of the null-space basis). *If $\mathbf{Z}$ and $\hat{\mathbf{Z}}$ have full column rank and the same range, then $\mathbf{Z}^\top \mathbf{H}_L \mathbf{Z}$ is positive semidefinite, respectively positive*

definite, if and only if $\widehat{\mathbf{Z}}^\top \mathbf{H}_L \widehat{\mathbf{Z}}$ has the same property.

Proof. There is a nonsingular square matrix $\mathbf{R}$ with $\widehat{\mathbf{Z}} = \mathbf{Z}\mathbf{R}$. Hence

$$\widehat{\mathbf{Z}}^\top \mathbf{H}_L \widehat{\mathbf{Z}} = \mathbf{R}^\top (\mathbf{Z}^\top \mathbf{H}_L \mathbf{Z}) \mathbf{R}.$$

For every $\mathbf{y}$, the quadratic form on the left equals the quadratic form of $\mathbf{Z}^\top \mathbf{H}_L \mathbf{Z}$ at $\mathbf{R}\mathbf{y}$. Since $\mathbf{R}$ is bijective, nonnegativity and strict positivity on nonzero vectors are preserved. $\square$

Example 13.3 (SONC filters a candidate through a reduced Hessian). Consider the three-dimensional problem

$$\min_{\mathbf{x} \in \mathbb{R}^3} f(\mathbf{x}) = x_1^2 - x_2^2 + x_3^2 \quad \text{subject to} \quad g(\mathbf{x}) = x_1 = 0.$$

At $\mathbf{x}^* = \mathbf{0}$, LICQ holds because $\nabla g(\mathbf{0}) = (1, 0, 0)^\top \neq \mathbf{0}$. Since $\nabla f(\mathbf{0}) = \mathbf{0}$, the unique KKT multiplier is $\lambda^* = 0$. Thus

$$C_{\text{crit}}(\mathbf{0}) = \{(0, d_2, d_3)^\top : d_2, d_3 \in \mathbb{R}\}, \quad \mathbf{H}_L = \text{diag}(2, -2, 2).$$

A basis and its reduced Hessian are

$$\mathbf{Z} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{Z}^\top \mathbf{H}_L \mathbf{Z} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix} \not\geq 0.$$

The critical direction $\mathbf{d} = (0, 1, 0)^\top$ has $\mathbf{d}^\top \mathbf{H}_L \mathbf{d} = -2 < 0$. SONC therefore rejects the origin as a local minimizer. Directly, the feasible restriction is $-x_2^2 + x_3^2$, which takes both signs near the origin.

Example 13.4 (SONC filters a candidate on a genuine cone). Change only the sign of the quadratic term in [Example 13.1](#):

$$\min_{x_1, x_2} f(\mathbf{x}) = x_1 - x_2^2 \quad \text{subject to} \quad h_1(\mathbf{x}) = -x_1 \leq 0, \quad h_2(\mathbf{x}) = -x_2 \leq 0.$$

At the origin, the active gradients $(-1, 0)^\top$ and $(0, -1)^\top$ are independent, so LICQ holds. The objective gradient is still $(1, 0)^\top$; hence the KKT first-order conditions still hold with $\boldsymbol{\nu}^* = (1, 0)^\top$, $J_+ = \{1\}$, and $J_0 = \{2\}$. Therefore

$$C_{\text{crit}}(\mathbf{0}) = \{(0, d_2)^\top : d_2 \geq 0\}, \quad \mathbf{H}_L = \begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix}.$$

The range of a matrix is always a subspace, so no matrix $\mathbf{Z}$ can have exactly this ray as its range. Test on the cone itself: $\mathbf{d} = (0, 1)^\top \in C_{\text{crit}}(\mathbf{0})$ gives

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} = -2 < 0.$$

SONC rejects the origin as a local minimizer, although it remains a KKT point. Indeed, the feasible points $(0, t)^\top$ satisfy $f(0, t) = -t^2 < f(\mathbf{0})$ for every $t > 0$.

Warning

Zero multipliers leave one-sided restrictions in the critical cone, so a subspace reduction is not generally available. Without LICQ, $J_0 \neq \emptyset$ does not by itself imply that the critical cone is not a subspace: some inequalities may be redundant after the homogeneous equalities have been imposed. For example, take $f(\mathbf{x}) = x_2^2$, $g(\mathbf{x}) = x_1 = 0$, and $h(\mathbf{x}) = -x_1 \leq 0$ at the origin. Choose the KKT multipliers $\lambda^* = \nu^* = 0$; then $J_0 = \{1\}$, yet $C_{\text{crit}}(\mathbf{0}) = \{(0, d_2)^\top : d_2 \in \mathbb{R}\}$: the inequality $-d_1 \leq 0$ adds no restriction once $d_1 = 0$ has been imposed.

13.5 KKT second-order sufficiency

SONC only imposes nonnegative critical curvature. Strengthen this to strict positivity on every nonzero critical direction. This stronger condition is designed as a sufficient certificate: directions in the critical cone are controlled by curvature, while the remaining linearized feasible directions already increase the objective to first order.

Definition 13.2 (Quadratic growth). A feasible point $\mathbf{x}^*$ satisfies quadratic growth if there are constants $c > 0$ and $\varepsilon > 0$ such that

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + c \|\mathbf{x} - \mathbf{x}^*\|^2 \quad (\mathbf{x} \in \mathcal{S}, \|\mathbf{x} - \mathbf{x}^*\| < \varepsilon). \quad (13.12)$$

Quadratic growth is stronger than strict local minimality: it gives a quantitative lower bound on the objective increase. The SOSC below guarantees precisely this property.

Lemma 13.10 (Uniform positive curvature on the critical cone). *Let $\mathbf{H}$ be symmetric and let C be a closed cone. The following statements are equivalent:*

1. $\mathbf{d}^\top \mathbf{H} \mathbf{d} > 0$ for every $\mathbf{d} \in C \setminus \{\mathbf{0}\}$;
2. there is $\eta > 0$ such that $\mathbf{d}^\top \mathbf{H} \mathbf{d} \geq \eta \|\mathbf{d}\|^2$ for every $\mathbf{d} \in C$.

Proof. The second statement immediately implies the first. Conversely, if $C = \{\mathbf{0}\}$, any $\eta > 0$ works. Otherwise, $C \cap \{\mathbf{d} : \|\mathbf{d}\| = 1\}$ is nonempty and compact because C is closed. The continuous function $\mathbf{d} \mapsto \mathbf{d}^\top \mathbf{H} \mathbf{d}$ has a strictly positive minimum η on this set. For nonzero $\mathbf{d} \in C$, apply the unit-vector bound to $\mathbf{d}/\|\mathbf{d}\| \in C$ and multiply by $\|\mathbf{d}\|^2$. The zero vector is immediate. $\square$

Since $C_{\text{crit}}(\mathbf{x}^*)$ is a closed cone, this lemma shows that the pointwise strict inequality in SOSC is equivalent to a uniform positive-curvature bound on the critical cone. The contradiction proof below uses the pointwise form directly.

Theorem 13.11 (KKT second-order sufficient condition (SOSC)). *Suppose the problem data are twice continuously differentiable and $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ satisfies the KKT first-order conditions in Definition 12.4. If*

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} > 0 \quad \text{for every } \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*) \setminus \{\mathbf{0}\}, \quad (13.13)$$

then $\mathbf{x}^$ satisfies quadratic growth. In particular, $\mathbf{x}^*$ is a strict local minimizer.*

Proof. Assume that quadratic growth fails. Then for each positive integer k there is a feasible point $\mathbf{x}^{(k)}$ such that

$$0 < \|\mathbf{x}^{(k)} - \mathbf{x}^*\| < \frac{1}{k}, \quad f(\mathbf{x}^{(k)}) < f(\mathbf{x}^*) + \frac{1}{k} \|\mathbf{x}^{(k)} - \mathbf{x}^*\|^2.$$

Set

$$t^{(k)} := \|\mathbf{x}^{(k)} - \mathbf{x}^*\|, \quad \mathbf{d}^{(k)} := \frac{\mathbf{x}^{(k)} - \mathbf{x}^*}{t^{(k)}}.$$

After passing to a subsequence, $\mathbf{d}^{(k)} \rightarrow \mathbf{d}$ with $\|\mathbf{d}\| = 1$.

First-order expansion of each equality gives $\nabla g_i(\mathbf{x}^*)^\top \mathbf{d} = 0$. If $j \in J(\mathbf{x}^*)$, feasibility and first-order expansion give $\nabla h_j(\mathbf{x}^*)^\top \mathbf{d} \leq 0$. Thus $\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*)$. Divide the objective inequality by $t^{(k)} > 0$ and use first-order expansion:

$$\nabla f(\mathbf{x}^*)^\top \mathbf{d} = \lim_{k \rightarrow \infty} \frac{f(\mathbf{x}^{(k)}) - f(\mathbf{x}^*)}{t^{(k)}} \leq 0.$$

KKT first-order stationarity Equation (12.7) and $\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*)$ give the reverse inequality through Equation (13.5). Hence $\nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0$ and $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$.

For every feasible $\mathbf{x}$, primal and dual feasibility Equations (12.8) and (12.9) give

$$L(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \leq f(\mathbf{x}),$$

because the equality terms vanish and $\nu_j^* h_j(\mathbf{x}) \leq 0$. Complementarity Equation (12.10) gives $L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = f(\mathbf{x}^*)$, and stationarity Equation (12.7) gives $\nabla_x L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{0}$. Therefore Taylor expansion of the Lagrangian at $\mathbf{x}^*$ yields

$$\frac{1}{2} (t^{(k)})^2 (\mathbf{d}^{(k)})^\top \mathbf{H}_L \mathbf{d}^{(k)} + o((t^{(k)})^2) \leq f(\mathbf{x}^{(k)}) - f(\mathbf{x}^*) < \frac{1}{k} (t^{(k)})^2.$$

Divide by $(t^{(k)})^2/2$ and let $k \rightarrow \infty$ to obtain $\mathbf{d}^\top \mathbf{H}_L \mathbf{d} \leq 0$. This contradicts Equation (13.13), because $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$ and $\|\mathbf{d}\| = 1$. Hence quadratic growth holds. $\square$

Important note: Why KKT SOSC does not require LICQ

The proof uses an actual sequence of feasible points and expands the Lagrangian directly. It never has to realize an arbitrary direction from $T_{\text{lin}}(\mathbf{x}^*)$ by a smooth feasible path. Therefore a constraint qualification is not needed once a KKT triple passing SOSC has been found. If multipliers are nonunique, it is enough that one specified KKT multiplier pair pass the test.

Corollary 13.12 (Reduced-Hessian, unconstrained, and zero-cone versions). *Under the hypotheses of Theorem 13.11:*

1. if $C_{\text{crit}}(\mathbf{x}^*) = \text{Range}(\mathbf{Z})$ is a subspace, then $\mathbf{Z}^\top \mathbf{H}_L \mathbf{Z} \succ 0$ implies quadratic growth and strict local minimality;
2. in the unconstrained case, the condition reduces to $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\mathbf{H}_f(\mathbf{x}^*) \succ 0$;
3. if $C_{\text{crit}}(\mathbf{x}^*) = \{\mathbf{0}\}$, SOSC holds vacuously and $\mathbf{x}^*$ satisfies quadratic growth.

Proof. The first claim follows from Proposition 13.8, which makes $\mathbf{Z}^\top \mathbf{H}_L \mathbf{Z} \succ 0$ equivalent to Equation (13.13). With no constraints, $C_{\text{crit}}(\mathbf{x}^*) = \mathbb{R}^n$ and $\mathbf{H}_L = \mathbf{H}_f(\mathbf{x}^*)$, giving the second claim. If the critical cone contains only $\mathbf{0}$, there are no nonzero critical directions to test, so Equation (13.13) is true vacuously. In every case, Theorem 13.11 yields quadratic growth. $\square$

In the zero-cone case, every nonzero linearized feasible direction is already strictly increasing to first order; no nonzero first-order-neutral direction remains for a curvature test.

13.5.1 Two successful applications of KKT SOSC

Example 13.5 (Curved equality). Return to [Example 13.2](#). At the origin,

$$C_{\text{crit}}(\mathbf{0}) = \{(d_1, 0)^\top : d_1 \in \mathbb{R}\}, \quad \mathbf{H}_L = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

For every nonzero critical direction,

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} = 2d_1^2 > 0.$$

SOSC gives quadratic growth and a strict local minimum. Notice that the objective Hessian alone is zero; the certificate comes entirely from the curved constraint.

Example 13.6 (An indefinite Hessian that is positive on a cone). Consider

$$\min_{x_1, x_2} f(\mathbf{x}) := \frac{1}{4}(x_1^2 + x_2^2) + x_1x_2 \quad \text{subject to} \quad -x_1 \leq 0, \quad -x_2 \leq 0. \quad (13.14)$$

At $\mathbf{x}^* = \mathbf{0}$, the KKT multipliers are $\boldsymbol{\nu}^* = \mathbf{0}$, so both active walls are unloaded and

$$C_{\text{crit}}(\mathbf{0}) = \mathbb{R}_+^2.$$

The Lagrangian Hessian is

$$\mathbf{H}_L = \begin{pmatrix} 1/2 & 1 \\ 1 & 1/2 \end{pmatrix},$$

whose eigenvalues are $3/2$ and $-1/2$. Thus it is indefinite in the ambient space. Nevertheless, for every nonzero $\mathbf{d} \in \mathbb{R}_+^2$,

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} = \frac{1}{2}(d_1^2 + d_2^2) + 2d_1d_2 > 0.$$

SOSC therefore certifies a strict local minimum. In fact every term in $f(\mathbf{x})$ is nonnegative on the feasible first quadrant, so the origin is the unique global minimizer.

13.6 Limits of the KKT second-order tests

13.6.1 The KKT second-order necessary condition is not sufficient

Example 13.7 (Nonnegative second-order curvature can miss an odd-order term). Consider

$$\min_{x_1, x_2} x_2^3 \quad \text{subject to} \quad x_1 = 0.$$

At the origin, LICQ holds, and the KKT first-order conditions hold with multiplier 0,

$$C_{\text{crit}}(\mathbf{0}) = \{(0, d_2)^\top : d_2 \in \mathbb{R}\}, \quad \mathbf{d}^\top \mathbf{H}_L \mathbf{d} = 0 \quad \text{for every } \mathbf{d} \in C_{\text{crit}}(\mathbf{0}).$$

Thus SONC holds. Nevertheless, the feasible restriction x_2^3 takes both signs arbitrarily close to zero, so the origin is not a local minimizer. The quadratic Taylor term is silent and the cubic term decides.

Warning

SONC is a necessary filter, not a certificate. Nonnegative critical curvature may contain nonzero zero-curvature directions, along which third- or higher-order terms determine the behavior.

13.6.2 The KKT second-order sufficient condition is not necessary

Example 13.8 (Strict minimality supplied by a fourth-order term). For

$$\min_{x_1, x_2} x_2^4 \quad \text{subject to} \quad x_1 = 0,$$

the origin is the strict global minimizer on the feasible line. At the origin,

$$C_{\text{crit}}(\mathbf{0}) = \{(0, d_2)^\top : d_2 \in \mathbb{R}\}, \quad \mathbf{d}^\top \mathbf{H}_L \mathbf{d} = 0 \quad \text{for every } \mathbf{d} \in C_{\text{crit}}(\mathbf{0}).$$

Hence SOSC fails even though strict global minimality holds. The first nonzero growth is fourth order.

Warning

SOSC is sufficient, so it cannot have a counterexample to sufficiency. The relevant limitation is that it is *not necessary*. Failure of SOSC is inconclusive; it may indicate a saddle or maximum, or it may simply mean that higher-order terms create the minimum.

Warning

Requiring $\mathbf{H}_L \succ 0$ on all of $\mathbb{R}^n$ is sufficient but generally far too strong. Only critical directions matter. Likewise, replacing every active inequality by an equality can omit inward directions associated with zero multipliers; when $J_0 \neq \emptyset$, the critical set may be a genuine cone rather than a subspace.

13.6.3 Necessary filter versus sufficient certificate

The two KKT second-order conditions have different hypotheses and different logical uses. The following table is a compact comparison; the complete proofs remain [Theorems 13.5](#) and [13.11](#).

Test	Hypotheses and curvature test	Logical conclusion
SONC	Assumptions: $\mathbf{x}^*$ is a local minimizer, the data are C^2 , and LICQ holds. Guaranteed curvature: $\mathbf{d}^\top \mathbf{H}_L \mathbf{d} \geq 0$ for every $\mathbf{d} \in C_{\text{crit}}$.	Necessary filter: one negative critical direction rejects the candidate under the stated hypotheses; passing SONC does not prove a minimum.
SOSC	Assumptions: the data are C^2 and the KKT first-order conditions hold. Required curvature: $\mathbf{d}^\top \mathbf{H}_L \mathbf{d} > 0$ for every $\mathbf{0} \neq \mathbf{d} \in C_{\text{crit}}$.	Sufficient certificate: quadratic growth and a strict local minimum; no constraint qualification is required.

13.7 A complete KKT first-order and second-order workflow

The diagram in Figure 13.4 combines the KKT first-order conditions with the KKT second-order necessary and sufficient conditions (SONC and SOSC). The dashed convex branch previews Chapter 14; there a *convex problem* means that f and every h_j are convex and every g_i is affine.

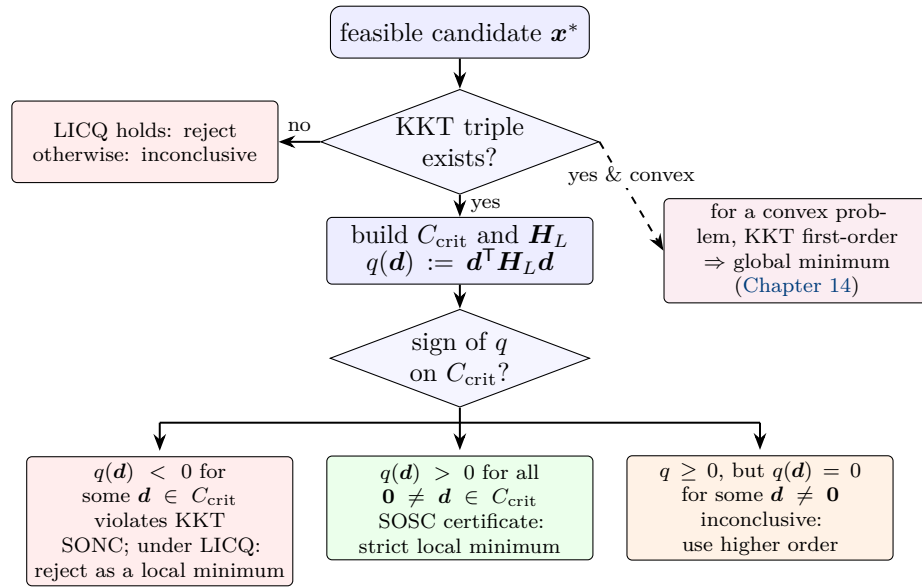


Figure 13.4: A KKT first-order and second-order workflow.

Important note: How to read the three curvature outcomes

- A negative critical direction violates SONC and rejects a candidate under the necessary-condition hypotheses.
- Strict positivity on all nonzero critical directions satisfies SOSC and proves quadratic growth without LICQ.
- Nonnegative curvature with a nonzero zero-curvature direction lies in the gap: inspect higher-order terms or restrict the objective directly to the feasible set.

Warning

Common mistakes in applying the KKT second-order conditions are:

1. testing $\mathbf{H}_f$ instead of the Lagrangian Hessian $\mathbf{H}_L$;
2. testing curvature on all of $\mathbb{R}^n$ instead of on $C_{\text{crit}}(\mathbf{x}^*)$;
3. treating every active inequality as an equality, including those with zero multiplier;
4. using an arbitrary feasible path in the SONC proof instead of a controlled path that keeps the loaded constraints active;
5. concluding that a point is not a minimum merely because SOSC fails;
6. forgetting that when LICQ fails, $\mathbf{H}_L$ may depend on which KKT multiplier pair is selected.

Chapter summary

- The critical cone consists of linearized feasible directions on which the first-order objective change vanishes. Loaded active inequalities become equalities in this cone, whereas unloaded active inequalities retain their one-sided restrictions.
- A controlled feasible curve shows that constraint curvature enters through the Lagrangian Hessian $\mathbf{H}_L$, rather than through the objective Hessian $\mathbf{H}_f$ alone.
- Under LICQ, the KKT second-order necessary condition (SONC) requires

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} \geq 0 \quad \text{for every } \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*).$$

Thus, one negative critical direction rejects a candidate under the necessary-condition hypotheses.

- The KKT second-order sufficient condition (SOSC) requires strict positivity on every nonzero critical direction. It guarantees quadratic growth and a strict local minimum, without requiring LICQ.
- When the critical cone is a subspace, a null-space basis produces a reduced Hessian that expresses the same curvature tests in lower-dimensional coordinates.
- Zero curvature along a nonzero critical direction is inconclusive: higher-order terms may still create or destroy local minimality.

13.8 Exercises

Exercise 13.1 (Critical cone from multipliers). For

$$\min(2x_1 + x_2) \quad \text{subject to} \quad -x_1 \leq 0, \quad -x_2 \leq 0,$$

verify the KKT multipliers at the origin, identify J_+ and J_0 , and compute the critical cone. Is the cone a subspace?

Exercise 13.2 (Equality-constrained classification). Classify the origin for

$$\min_{x_1, x_2} 2x_1^2 + 5x_1x_2 + x_2^2 \quad \text{subject to} \quad x_1 - 2x_2 = 0$$

using a reduced Hessian. Confirm the answer by eliminating the equality.

Exercise 13.3 (An unloaded active wall). For

$$\min_{x_1, x_2} x_1^4 + x_2^2 \quad \text{subject to} \quad -x_1 \leq 0,$$

compute the KKT multipliers and the critical cone at the origin. Determine whether SONC and SOSC hold, and then classify the point directly.

Exercise 13.4 (Constraint curvature). For

$$\min_{x_1, x_2} -2x_2 + ax_1^2 \quad \text{subject to} \quad x_2 - 3x_1^2 = 0,$$

derive the multiplier and Lagrangian Hessian at the origin. Determine for which values of a SONC or SOSC holds, and compare with the objective restricted to the feasible curve.

Exercise 13.5 (Reduced Hessian in three dimensions). Consider

$$\min_{\mathbf{x} \in \mathbb{R}^3} \frac{1}{2} \mathbf{x}^\top \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 4 \end{pmatrix} \mathbf{x} \quad \text{subject to} \quad x_1 + x_2 + x_3 = 0.$$

Construct a basis $\mathbf{Z}$ for the critical subspace, compute the reduced Hessian, and classify the origin. Repeat with a different basis and verify [Proposition 13.9](#) numerically.

Exercise 13.6 (A critical cone that is not a subspace). Construct a two-dimensional KKT triple with one loaded active wall and one unloaded active wall such that the critical cone is a genuine half-line. Choose a symmetric Lagrangian Hessian that satisfies SONC but not SOSC on this cone, and construct a smooth objective realizing your data.

Exercise 13.7 (Testing the regularity assumptions). For the feasible set $\{(x_1, x_2) : h(\mathbf{x}) := x_2^4 \leq 0\}$ at the origin, compute the tangent and linearized cones and decide whether LICQ holds. For

$$f(\mathbf{x}) := x_1^2 - x_2^2,$$

show that the origin is a strict constrained minimizer, while every KKT multiplier gives the same Lagrangian Hessian and testing it on the entire linearized cone finds a negative-curvature direction. Explain precisely why [Theorem 13.5](#) does not apply.